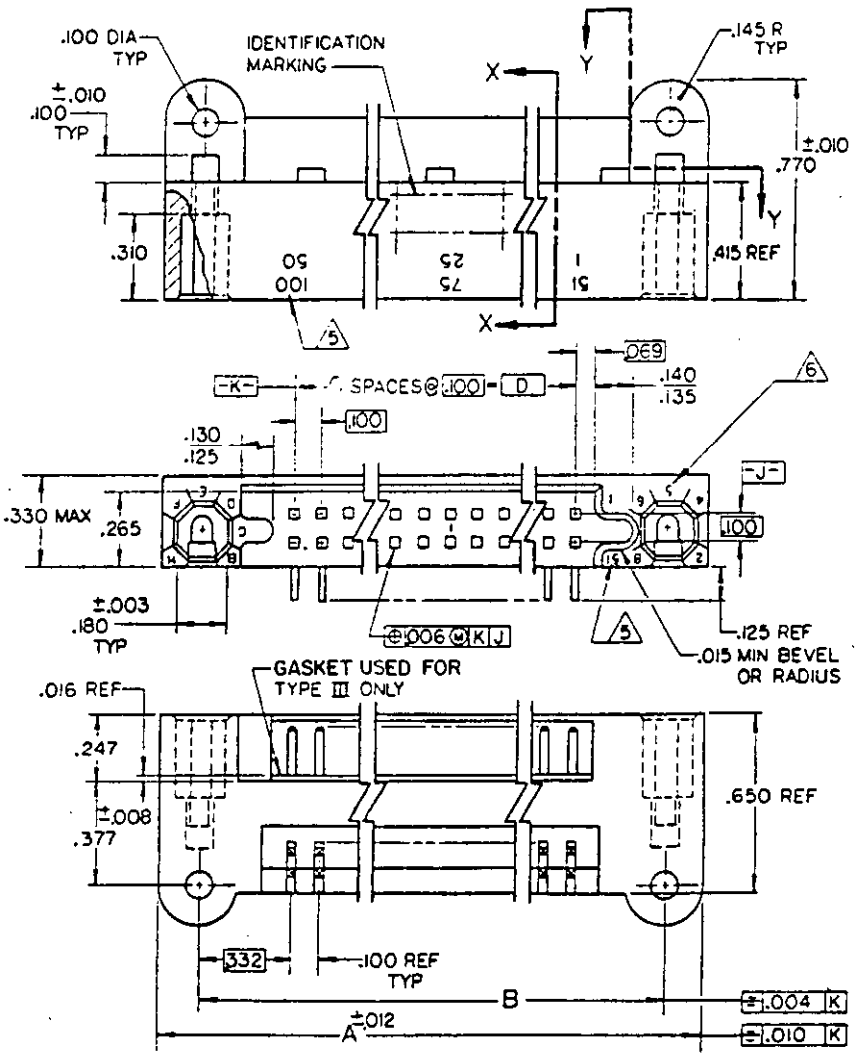
MILITARY SPECIFICATION SHEET

CONNECTORS, PRINTED CIRCUIT SUBASSEMBLY AND ACCESSORIES: PLUG, RIGHT ANGLE, 30 THRU 140 CONTACT POSITIONS, FOR PRINTED WIRING BOARDS (.100 SQ. GRID)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

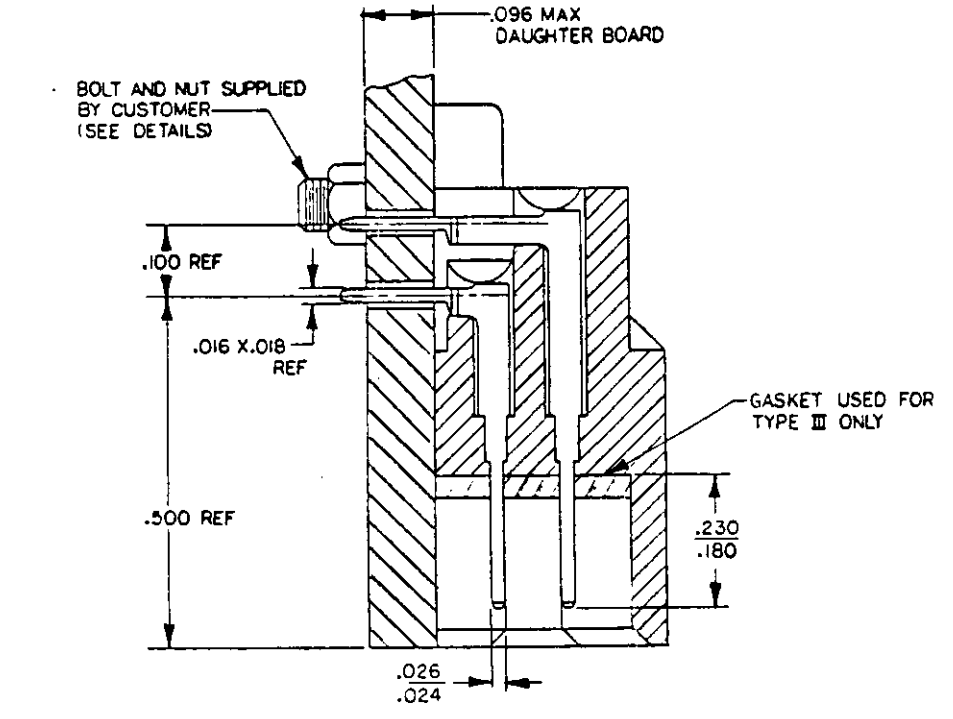
The complete requirements for acquiring the connectors described herein shall consist of this specification and the latest issue of MIL-C-55302.



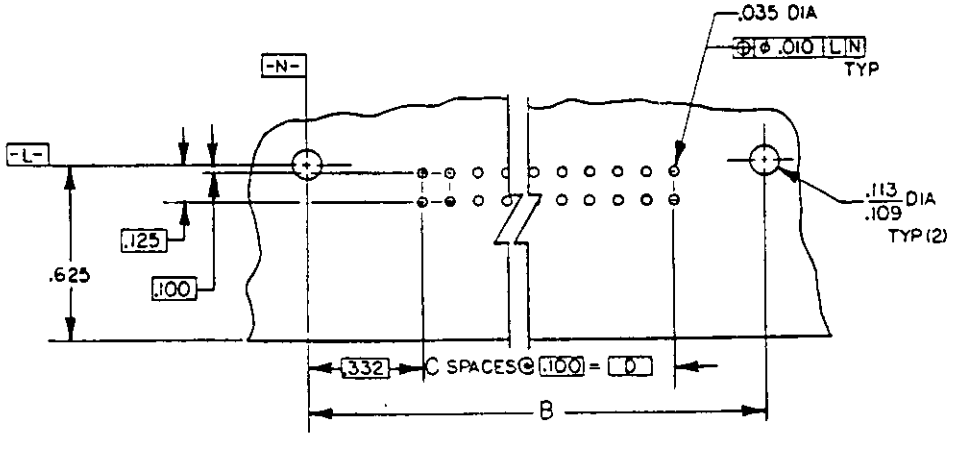
Ⓔ FIGURE 1. Connectors, plug (.100 sq. grid).

FSC 5935

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T22:50Z
Check the source to verify that this is the current version before use.



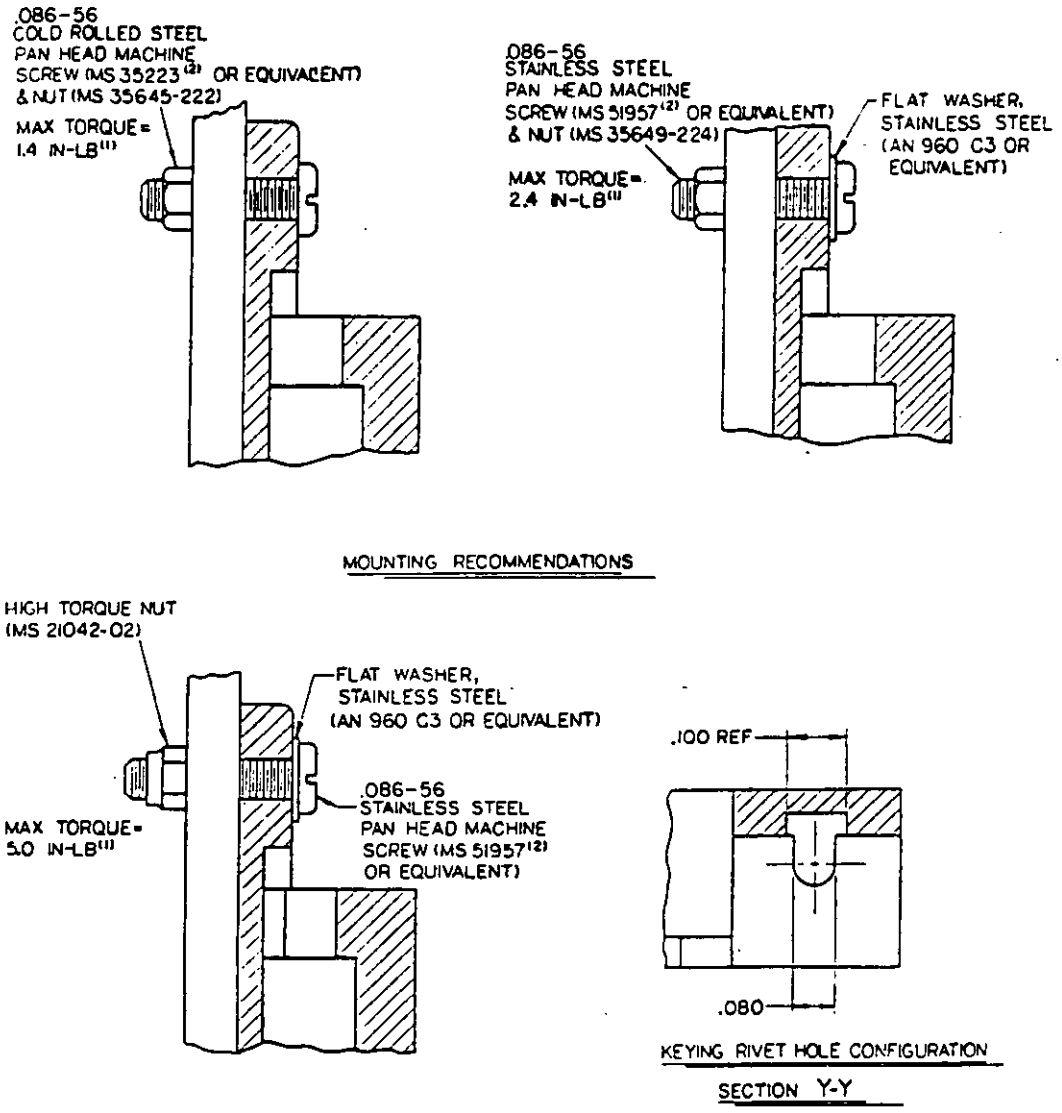
SECTION X-X



RECOMMENDED P.C. BOARD LAYOUT

Ⓔ FIGURE 1. Connectors, plug (.100 square grid) for types II and III - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T22:50Z
Check the source to verify that this is the current version before use.



NOTES:
1. Torque wrench must be used when fastening connector to printed circuit board.
2. Length to be determined by user.

Ⓔ FIGURE 1. Connectors, plug (.100 square grid) - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T22:50Z
Check the source to verify that this is the current version before use.

Dash No. 2/	Type No.	No. of contacts	Dimensions 1/ A	B	C	D	Contact identification numbers
10 19	II III	30	2.355 (59.82)	2.065 (52.45)	14	1.400 (35.56)	1, 16
11 20	II III	40	2.855 (72.52)	2.565 (65.15)	19	1.900 (48.26)	1, 21
12 21	II III	50	3.355 (85.22)	3.065 (77.85)	24	2.400 (60.96)	1, 26
13 22	II III	60	3.855 (97.92)	3.565 (90.55)	29	2.900 (73.66)	1, 31
14 23	II III	70	4.355 (110.62)	4.065 (103.25)	34	3.400 (86.36)	1, 36
15 24	II III	80	4.855 (123.32)	4.565 (115.95)	39	3.900 (99.06)	1, 41
16 25	II III	90	5.355 (136.02)	5.065 (128.65)	44	4.400 (111.76)	1, 46
17 26	II III	100	5.855 (148.72)	5.565 (141.35)	49	4.900 (124.46)	1, 51
18 27	II III	110	6.355 (161.42)	6.065 (154.05)	54	5.400 (137.16)	1, 56
28 32	II III	120	6.855 (174.12)	6.565 (156.75)	59	5.900 (149.86)	1, 61
29 33	II III	130	7.355 (186.82)	7.065 (158.44)	64	6.400 (162.56)	1, 66
30 34	II III	134	7.555 (191.90)	7.265 (184.53)	66	6.600 (167.64)	1, 68
31 35	II III	140	7.855 (199.52)	7.565 (192.15)	69	6.900 (175.26)	1, 71

FIGURE 1. Connectors, plug (.100 square grid) for types II and III - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T22:50Z
Check the source to verify that this is the current version before use.

INCHES	MM	INCHES	MM	INCHES	MM
.003	0.08	.058	1.47	.140	3.56
.005	0.13	.069	1.75	.180	4.57
.006	0.15	.072	1.83	.230	5.84
.008	0.20	.096	2.44	.247	6.27
.010	0.25	.100	2.54	.265	6.73
.012	0.30	.101	2.57	.310	7.87
.015	0.38	.109	2.77	.323	8.20
.024	0.61	.113	2.87	.332	8.43
.025	0.64	.125	3.18	.377	9.58
.026	0.66	.130	3.30	.415	10.54
.030	0.76	.135	3.43	.500	12.70
				.770	19.56

1/ Metric equivalents are in parentheses.
2/ Type I listed as -01 through -09 in previous revision C dated March 24, 1980 are in active for new design, type II listed as -10 through -18 are replacements for type I respectively.

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±.005 (0.13 mm) on decimals and ±2° on angles.
4. These connectors mate with connectors specified in MIL-C-55302/27.
5. Numbers indicating every five cavities stamped on side. Numbers indicating end cavities and line indicating every five cavities stamped on mating face.
6. Key locations embossed on indicated surfaces.

FIGURE 1. Connectors, plug (.100 square grid) for types II and III - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T22:50Z
Check the source to verify that this is the current version before use.

REQUIREMENTS

Design and construction:

Dimensions and configuration: See figure 1 and table I.

Material: In accordance with MIL-C-55302.

Gasket: Silicon rubber.

Ⓔ Plating: The contact plating for the engagement area, .220 inches minimum length, shall be gold per MIL-G-45204, type II, grade C, class 1, over nickel per QQ-N-290, class 2, 50 to 150 microinches. The contact plating for the solder tail area, .160 inches minimum length, shall be tin lead per MIL-P-81728, 100 microinches minimum over nickel per QQ-N-290, class 2, 50 to 150 microinches.

Contact identification: See figure 1.

Contact rating: 3.0 amperes maximum per contact, 2.25 amperes continuous in accordance with connector at room ambient.

Ⓔ Keying: See MIL-C-55302/31, using the M55302/31-10 rivet.

Mating and unmating: The maximum mating force in pounds shall be the number of contacts multiplied by .25 and the withdrawal force in pounds shall be a minimum of .025 times the number of contacts and shall not exceed the measured insertion force.

Contact resistance: No individual contact pair shall have a resistance exceeding .020 ohm.

Contact retention: 3 pounds minimum.

Dielectric withstanding voltage:

Sea level: 900 volts rms.

High altitude: 200 volts rms.

Part number: M55302/110- (and dash number from figure 1).

The Government has a royalty free license under the following patent for the benefit of manufacturers of the item either for the Government or for use in equipment to be delivered to the Government.

US Patent number 3,404,367

Custodians:
Army - CR
Navy - EC
Air Force - 17

Review activities:
Army - CR, AR, MI
Navy - AS, OS
Air Force - 11, 99
DLA - ES

User activities:
Army - ME, AT, AV
Navy - MC
Air Force - 19

Agent:
DLA - ES

Preparing activity:
Army - CR

(Project 5935-3379)

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T22:50Z
Check the source to verify that this is the current version before use.